Scalability & Future Plan

Date	15 March 2026
Team ID	0827CY231062
Project Name	Health360: Executive Analytics & Population Health Dashboard
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No.	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static Flat-File Data Ingestion	Data refreshes require manual CSV file updates and extract regeneration rather than automated real-time streaming.	High
2	Synthetic Cohort Sample Size	The current 5,000-patient dataset does not fully capture rare clinical edge cases or complex multi-morbidity patterns.	Medium
3	Rule-Based Risk Calculation Engine	The composite Calculated Risk Score relies on static weighted logic rather than dynamic predictive machine learning models.	High

Scalability Plan

S. No.	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Public web access via Tableau Public Server.	Migrate to **Tableau Server / Cloud Containerized Infrastructure** with Role-Based Access Control (RBAC) to support thousands of concurrent clinical users.

2	Data Storage	Local flat CSV file (smart_healthcare_dataset.csv) converted to .hyper extract.	Connect to enterprise cloud data warehouses like **Snowflake**, **Amazon Redshift**, or **Google BigQuery** for multi-terabyte EHR storage.
3	Performance	In-memory Tableau extract processing.	Implement **Incremental Data Extract Refreshing** and query optimization partitioning to handle millions of patient records with sub-second latency.
4	Security	Synthetic, fully anonymized cohort without PII.	Implement **HIPAA-compliant row-level security (RLS)** and end-to-end data encryption at rest and in transit for sensitive patient records.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud Data Pipeline & Live ETL	Q2 2026	Automate daily/hourly data ingestion from cloud databases (e.g., Snowflake) via automated ETL pipelines.
Phase 2	Predictive AI/ML Risk Modelling	Q2 2026	Integrate Python (TabPy) predictive models to forecast 5-year chronic disease onset and readmission probabilities.
Phase 2	Automated Clinical Alerting Engine	Q2 2026	Deploy automated email/SMS alerts to clinical care teams when patient risk scores breach critical thresholds.